Department of Economics
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# ECON7335: Problem Set 1

1. **Economy 1 with endogenous information**: Consider a static economy in which agents' goal is to track an exogenous fundamental $\theta$. That is, each agent's action is given by

$$a_i = E^i[\theta]. \tag{1}$$

Each agent's information set, $\Omega^i$, consists of an exogenous, idiosyncratic signal

$$s_i = \theta + v_i \tag{2}$$

and an endogenous, aggregate signal

$$g = a + \eta. \tag{3}$$

The noise terms are distributed normally, with mean zero and variances $\sigma_v^2$ and $\sigma_\eta^2$ respectively. Moreover, the shocks $v_i$ are uncorrelated across agents.

(a) Conjecture an equilibrium relationship for aggregate action (call it the PLM)

$$a = \theta + \phi_2 \eta. \tag{4}$$

Derive the optimal inference of agent $i$ given the conjecture.

(b) Using your answer in (a), solve the equilibrium low of motion (the ALM) for the aggregate action, given the conjectured PLM.

(c) Show that the relationship you derived in (b) implies that the equilibrium coefficient $\phi_2$ is the solution to a third-order equation. (But DON'T solve for it!)

2. **Economy 2 with endogenous information:** Consider a static economy in which agents' goal is to track an exogenous fundamental $\theta$, with losses from missing the target normalized by their variance. That is, each agent's action is given by

$$a_i = E^i[\theta]/Var(\theta|\Omega^i). \tag{5}$$

Each agent's information set, $\Omega^i$, consists of an exogenous, idiosyncratic signal

$$s_i = \theta + v_i \tag{6}$$

and an endogenous, aggregate signal

$$g = a + \eta. \tag{7}$$

The noise terms are distributed normally, with mean zero and variances $\sigma_v^2$ and $\sigma_\eta^2$ respectively. Moreover, the shocks $v_i$ are uncorrelated across agents. (The only change between the earlier equation is the rescaling of the optimal action.)

(a) Conjecture an equilibrium relationship for aggregate action (call it the PLM)

$$a = \phi_1\theta + \phi_2\eta. \tag{8}$$

Derive the optimal inference of agent $i$ given the conjecture.

(b) Using your answer in (a), solve the equilibrium low of motion (the ALM) for the aggregate action, given the conjectured PLM.

(c) Show that the relationship you derived in (b) implies that the equilibrium coefficient $\phi_2$ is the solution to a linear equation. (And, please do solve for it!)

(d) Can you provide any intuition for the different form of the results in 1.c and 2.c?